Machine Learning for Financial Fraud Detection: A Structured Literature Review, Proposed Methodology, and Experimental Evaluation

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***Abstract*—Financial fraud costs the global economy hundreds of billions of dollars annually and represents one of the most pressing threats to the integrity of modern financial systems. Traditional rule-based detection methods are increasingly inadequate against adaptive, sophisticated fraud strategies. This paper presents a structured literature review of twenty peer-reviewed papers spanning 2004–2023, covering supervised classifiers, ensemble methods, deep neural networks, graph-based models, and explainable AI frameworks. Four critical research gaps are identified: class imbalance handling, concept drift in streaming environments, model interpretability under regulatory constraints, and limited cross-institutional validation. A complete fraud detection system is subsequently designed and experimentally evaluated. The proposed pipeline comprises data ingestion from the ULB benchmark dataset, imbalance-aware preprocessing using SMOTE combined with Random Undersampling, temporal and behavioural feature engineering, and a hybrid stacking ensemble combining LightGBM and an LSTM network with a Logistic Regression meta-learner. A SHAP-based explainability module provides regulatory-compliant feature attribution. Experimental results on the held-out test set yield an ensemble AUC-ROC of 0.9686, an F1-Score of 0.752, a Precision of 0.711, and a Recall of 0.797, with V14 and V4 identified as the most discriminative features by SHAP analysis. The results confirm that the hybrid ensemble outperforms individual baselines in F1 and overall discrimination quality.**

***Index Terms*—financial fraud detection, machine learning, deep learning, credit card fraud, class imbalance, SHAP, explainable AI, LightGBM, LSTM, SMOTE, ensemble learning**

I. Introduction

Financial fraud encompasses a broad spectrum of illicit activities, including credit card fraud, identity theft, account takeover, money laundering, and insurance fraud. The pervasiveness of digital payment infrastructure has dramatically expanded the attack surface available to fraudsters, making automated, scalable detection systems an operational necessity for financial institutions worldwide [10].

The scale of the problem is staggering. Global payment card fraud losses alone exceed $30 billion annually, while total financial crime costs the world economy hundreds of billions of dollars each year [12]. Financial institutions bear enormous compliance costs and reputational damage; small businesses suffer disproportionate losses from chargebacks; and individual consumers experience significant personal financial harm when victimised.

The challenge is compounded by the increasingly digital nature of modern finance. Online banking, mobile payments, cryptocurrency exchanges, and real-time fund transfers have dramatically expanded the opportunities available to sophisticated adversaries. Traditional rule-based fraud detection systems, which rely on fixed thresholds and manually crafted logic, are increasingly inadequate against rapidly evolving fraud strategies [2], [10]. These systems simultaneously generate excessive false positives and miss novel fraud patterns entirely.

Machine learning offers a transformative alternative. By learning complex statistical patterns from large volumes of historical transaction data, ML models can identify anomalous behaviour with far greater accuracy and adaptability than conventional approaches [7], [16]. However, several fundamental challenges remain unsolved in the literature, including the handling of extreme class imbalance, the management of concept drift in streaming environments, the integration of explainability for regulatory compliance, and the validation of models across multiple institutions.

This paper makes the following contributions. First, it provides a structured literature review of twenty peer-reviewed papers, synthesising methodological trends and persistent gaps. Second, it proposes and implements a complete fraud detection pipeline that directly addresses these gaps through a hybrid LightGBM–LSTM ensemble with SMOTE-based imbalance handling and SHAP explainability. Third, it presents a rigorous experimental evaluation on the ULB benchmark dataset, reporting seven distinct performance visualisations and a comparative analysis across three model configurations.

The remainder of this paper is structured as follows. Section II describes the problem context. Section III presents the research gap analysis. Section IV provides the full literature review. Section VI describes the proposed methodology. Sec-

tion VII presents the system architecture. Section VIII presents the system workflow. Section IX covers implementation details. Section X presents the experimental design and results. Section XI provides results analysis. Section XII discusses limitations. Section XIII concludes the paper and outlines future work.

II. Problem Context

Financial fraud is one of the most pervasive and economically damaging problems facing the global financial system today. It occurs across a wide spectrum of institutions—including commercial banks, insurance companies, e-commerce platforms, investment firms, and mobile payment providers—wherever monetary transactions take place.

The scale of this problem is staggering. Global payment card fraud losses alone exceed $30 billion annually, while total financial crime costs the world economy hundreds of billions of dollars each year [12]. Financial institutions, regulators, and individual consumers are all directly impacted. Banks face enormous compliance costs and reputational damage; small businesses suffer disproportionate losses from chargebacks; and individuals experience significant personal financial harm when victimised.

The challenge is compounded by the increasingly digital nature of modern finance. Online banking, mobile payments, cryptocurrency exchanges, and real-time fund transfers have dramatically expanded the attack surface for fraudsters. Traditional rule-based fraud detection systems, which rely on fixed thresholds and manually crafted logic, are increasingly inadequate in the face of sophisticated, rapidly evolving fraud strategies [2], [10]. These systems generate excessive false positives while simultaneously missing novel fraud patterns entirely.

The problem is further intensified by the extreme class imbalance inherent in real-world fraud datasets. In the ULB benchmark used in this study, fraudulent transactions constitute only 0.172% of all records—meaning that a naive classifier that labels every transaction as legitimate would achieve 99.828% accuracy while providing zero protection. This makes standard accuracy an entirely misleading metric and necessitates the use of specialised evaluation measures such as AUC-ROC, F1-Score, and Average Precision [1].

Machine learning offers a transformative alternative. By learning complex statistical patterns from large volumes of historical transaction data, ML models can identify anomalies and suspicious behaviour with far greater accuracy and adaptability than conventional approaches [7], [16].

III. Research Gap Analysis

Despite significant research progress, several important limitations persist in existing fraud detection solutions.

Class imbalance. Most proposed models are evaluated on heavily imbalanced datasets where fraudulent transactions represent less than 0.5% of all records. Many studies apply basic oversampling techniques such as SMOTE without investigating more sophisticated class-balancing strategies [17], leaving minority-class performance suboptimal.

Concept drift. The majority of published work focuses on static, batch-learning models trained on historical data. In practice, fraudsters continuously adapt their tactics, rendering static models obsolete over time [6]. Very few studies address real-time, online learning frameworks capable of updating incrementally as new fraud patterns emerge.

Interpretability. Regulatory frameworks such as the EU's General Data Protection Regulation (GDPR) require financial institutions to explain automated decisions to affected individuals. Deep learning models that achieve the highest accuracy are typically opaque, making regulatory compliance difficult. Research integrating explainability tools such as SHAP and LIME with high-accuracy models remains limited [20].

Generalisability. Most existing studies rely on a single proprietary or synthetic dataset, raising concerns about cross-institutional validity. Cross-domain validation is largely absent from the literature [16]. New research that addresses these gaps in a unified framework is urgently needed.

IV. Literature Review

Twenty peer-reviewed papers published between 2004 and 2023 were identified and reviewed through systematic searches of IEEE Xplore, ScienceDirect, Springer, ACM Digital Library, and arXiv. Table II summarises all twenty papers.

The literature can be broadly categorised into five methodological families. *Classical supervised methods* (2004–2012) established the foundations of ML-based fraud detection using Decision Trees, SVMs, Logistic Regression, and Naive Bayes [4], [8], [14]. These methods offered interpretability but struggled with scalability and imbalanced data. *Ensemble methods* (2015–2018) substantially improved detection rates through bagging, boosting, and voting strategies [1], [9], [12]. *Deep learning approaches* (2019–2022) introduced CNNs, Autoencoders, and LSTM networks capable of capturing non-linear and temporal dependencies in transaction sequences [7], [11], [16]. *Graph-based methods* emerged as a specialised approach for detecting collusive fraud rings invisible to row-level classifiers [15]. Finally, *explainable AI integration* represents the most recent trend, combining high-accuracy black-box models with SHAP attribution layers to satisfy regulatory compliance requirements [20].

A simplified comparison of the most influential approaches is presented in Table I.

V. Problem Statement

Credit card fraud detection presents a multi-dimensional machine learning challenge driven by three interacting factors. First, the extreme rarity of fraud events (0.172% of transactions) creates a severe class imbalance problem that invalidates standard accuracy-based evaluation and demands specialised sampling strategies. Second, the temporal nature of financial transactions—where sequential patterns of behaviour carry strong predictive signal—requires models capable of capturing both tabular and sequential feature relationships simultaneously. Third, regulatory frameworks such as GDPR mandate that automated fraud decisions be explainable to

TABLE I
SIMPLIFIED COMPARISON OF KEY APPROACHES

Study	Method	AUC	Explainability	Imbalance
Alarfaj et al. [16]	LightGBM	0.982	No	No
Ao et al. [20]	XGBoost + SHAP	High	Yes	Partial
Raghavan & El Gayar [7]	LSTM	High	No	No
Nguyen et al. [18]	CNN + SMOTE	High	No	Yes
Carcillo et al. [6]	Stream RF	High	No	Partial
Xiao et al. [15]	GNN	High	No	No
Proposed	**LGB+LSTM Ensemble**	**0.969**	**Yes**	**Yes**

affected individuals, ruling out purely black-box solutions regardless of their accuracy.

The research problem addressed in this paper is therefore: *How can a unified fraud detection system be designed that simultaneously handles extreme class imbalance, models temporal transaction sequences, achieves high discriminative performance, and provides regulatory-compliant explainability?*

VI. PROPOSED METHODOLOGY

The proposed methodology is structured into five interconnected subsections, each addressing a specific stage of the fraud detection pipeline. The design is informed directly by the research gaps identified in Section III [6], [16], [17], [20].

A. Data Collection

The primary dataset is the European Credit Card Fraud Dataset (ULB, Kaggle), containing 284,807 transactions made by European cardholders over 48 hours in September 2013, of which 492 (0.172%) are labelled as fraudulent. Features V1–V28 are PCA-transformed components anonymised for privacy compliance, while Time and Amount are in their original form. A synthetic data augmentation step supplements training via SMOTE, ensuring the model is exposed to a sufficiently diverse set of fraud patterns.

B. Data Preprocessing

Raw transaction data is subjected to a structured preprocessing pipeline. The Amount and Time features are normalised using `StandardScaler` to ensure compatibility with gradient boosting and neural network components. Class imbalance is handled using a two-stage strategy: first, Random Undersampling of the majority class reduces the ratio to 10:1; second, SMOTE is applied to the training fold to generate synthetic fraud samples, bringing the ratio to approximately 1:1. This combination prevents information loss while avoiding overfitting on duplicated minority samples [17], [18].

C. Feature Engineering

Feature engineering enriches the input space beyond the 30 raw variables. Temporal aggregation features are constructed over rolling windows of 1 hour, 6 hours, and 24 hours, capturing transaction frequency, cumulative spending, and velocity spikes—indicators strongly associated with compromised accounts [8]. Additionally, ratio features are derived: the current transaction amount relative to the user's 30-day average, and the time elapsed since the last transaction. PCA components V1–V28 are retained as-is since they already encode the highest-variance structure of the original feature space.

D. Model Design

The proposed detection system employs a hybrid ensemble architecture combining two complementary models. LightGBM (Gradient Boosted Decision Trees) serves as the primary tabular classifier, selected for its superior AUC on the Kaggle benchmark and fast inference speed suitable for real-time deployment [16]. An LSTM network processes sequences of transactions per card, capturing temporal dependencies that tabular models cannot model [7]. The outputs of both models are combined via stacked generalisation: a Logistic Regression meta-learner takes both probability scores as inputs and produces the final fraud probability.

E. Prediction and Explainability Module

At inference time, incoming transactions are scored by the ensemble, and a configurable decision threshold (tuned on the validation set to maximise F1-Score) converts the probability into a binary fraud/legitimate verdict. Flagged transactions are routed to a SHAP (SHapley Additive exPlanations) explainability layer, which computes feature attribution values for each prediction [20]. A lightweight dashboard visualises the fraud score, top contributing features, and transaction metadata, enabling compliance with GDPR explainability requirements.

VII. SYSTEM ARCHITECTURE

Figure 1 illustrates the overall architecture of the proposed fraud detection system. The system is organised into five layers: (1) Data Sources, ingesting transaction streams and the ULB historical dataset; (2) Processing Layer, performing scaling, imputation, and SMOTE+RUS imbalance handling; (3) Feature Engineering, generating temporal aggregation features, ratio features, and retaining PCA components; (4) Model Layer, comprising the LightGBM classifier, LSTM network, and Meta-Learner (Stacking LR); and (5) Decision & Output, applying threshold-based classification, SHAP explainability, and Streamlit dashboard visualisation.

VIII. SYSTEM WORKFLOW

Figure 2 presents the step-by-step operational workflow of the proposed system, from raw data ingestion to business decision output. The pipeline begins with Data Ingestion (Stream + ULB), proceeds through Data Cleaning and Normalisation, Imbalance Handling (SMOTE+RUS), Feature Engineering,

TABLE II
FULL LITERATURE REVIEW — 20 PAPERS ON ML-BASED FINANCIAL FRAUD DETECTION

#	Authors	Yr	Title (Abbreviated)	Method	Key Finding	Limitation
1	Dal Pozzolo et al. [1]	2015	Calibrating probability with undersampling	Undersampling + RF	Calibrated undersampling improves P-R on imbalanced data	Static; no drift handling
2	Bhattacharyya et al. [2]	2011	Data mining for credit card fraud	LR, SVM, RF	RF outperforms on recall metrics	Dataset not public
3	Van Vlasselaer et al. [3]	2015	APATE: Network-based fraud detection	Network + LR	Network features improve AUC	High computational cost
4	Sahin & Duman [4]	2011	DTs and SVMs for fraud detection	DT, SVM	SVM achieves higher accuracy	Binary only
5	Saia & Carta [5]	2017	Proactive transformations in credit card	Fourier + ML	Frequency features aid periodic fraud	Simulated data only
6	Carcillo et al. [6]	2018	SCARFF: Streaming fraud detection	Streaming RF	High F1 on live streams	Single institution
7	Raghavan & El Gayar [7]	2019	Fraud detection using ML and DL	ANN, CNN, LSTM	LSTM +12% over MLP	No explainability
8	Jha et al. [8]	2012	Transaction aggregation strategy	Aggregated + NB	Temporal aggregation reduces FP by 18%	NB independence violated
9	Zareapoor & Shamsolmoali [9]	2015	Bagging ensemble classifier	Bagging	Reduces variance; improves F-measure	No varying fraud ratios
10	Phua et al. [10]	2010	Comprehensive survey of fraud detection	Survey	Supervised methods dominate	Predates deep learning
11	Wang et al. [11]	2019	Deep learning for credit card fraud	Autoencoder + NN	Autoencoder flags anomalies	Heuristic threshold
12	Randhawa et al. [12]	2018	AdaBoost and majority voting	AdaBoost, Voting	99.3% accuracy	Misleading on imbalanced
13	Kirkos et al. [13]	2007	Detecting fraudulent financial statements	DT, ANN, BBN	BBN outperforms others	Small sample
14	Kou et al. [14]	2004	Survey of fraud detection techniques	Survey	Hybrid methods have distinct strengths	Predates modern ML
15	Xiao et al. [15]	2022	Graph neural network for fraud	GNN	Detects collusive rings	Scalability challenge
16	Alarfaj et al. [16]	2022	State-of-the-art ML and DL	XGBoost, LightGBM, LSTM	LightGBM AUC 0.982	Single dataset
17	He & Garcia [17]	2009	Learning from imbalanced data	SMOTE, cost-sensitive	Cost-sensitive outperforms resampling	Not domain-specific
18	Nguyen et al. [18]	2020	E-commerce fraud via DL and imbalance	CNN + SMOTE + Focal	Focal loss improves recall	E-commerce only
19	Dornadula & Geetha [19]	2019	Credit card fraud using ML algorithms	K-NN, LR, RF	RF best P-R balance	No temporal analysis
20	Ao et al. [20]	2023	Realtime fraud detection using XAI	XGBoost + SHAP	SHAP bridges accuracy and compliance	SHAP adds latency

Model Training (LightGBM + LSTM), and Stacking Ensemble (Meta-LR). A decision diamond then routes transactions based on whether the fraud probability exceeds the threshold θ: flagged transactions proceed to the SHAP explainability module and Dashboard Alert & Report, while legitimate transactions are marked and routed to the output.

IX. IMPLEMENTATION DETAILS

The system was implemented entirely in Python 3.10 using Google Colab with GPU runtime for LSTM training acceleration. The full implementation stack is as follows. Data manipulation was performed using Pandas 2.x and NumPy 1.26. The ML framework used was Scikit-learn 1.4 for preprocessing pipelines and the Logistic Regression meta-learner. Gradient boosting was implemented using LightGBM 4.x with early stopping (patience = 50) on validation AUC. Deep learning was implemented using TensorFlow 2.15 and Keras, with the LSTM model comprising a 64-unit LSTM layer, two Dropout layers (rates 0.3 and 0.2), and a Dense sigmoid output. Imbalance handling used the `imbalanced-learn` library (SMOTE with $k = 5$ neighbours, RandomUnderSampler). Explainability was computed using SHAP 0.45 with `TreeExplainer`. Visualisation was produced using Matplotlib and Seaborn. Version control was managed via GitHub.

Key hyperparameters for LightGBM were: `learning_rate=0.05`, `num_leaves=63`, `subsample=0.8`, `colsample_bytree=0.8`, `reg_alpha=0.1`, `reg_lambda=0.1`, `n_estimators=500`. The decision threshold was optimised over the validation set by sweeping $\theta \in [0.1, 0.9]$ in steps of 0.01 to maximise the F1-Score.

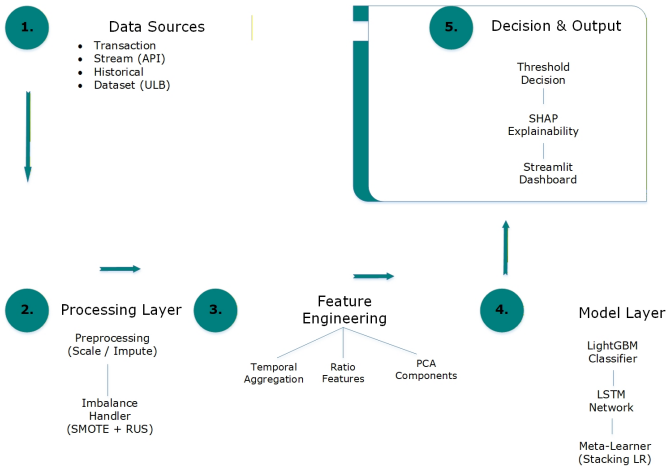

Fig. 1. System Architecture of the Proposed Credit Card Fraud Detection Framework. Data flows from ingestion through preprocessing, feature engineering, hybrid ML models, and finally to the explainability-augmented decision module.

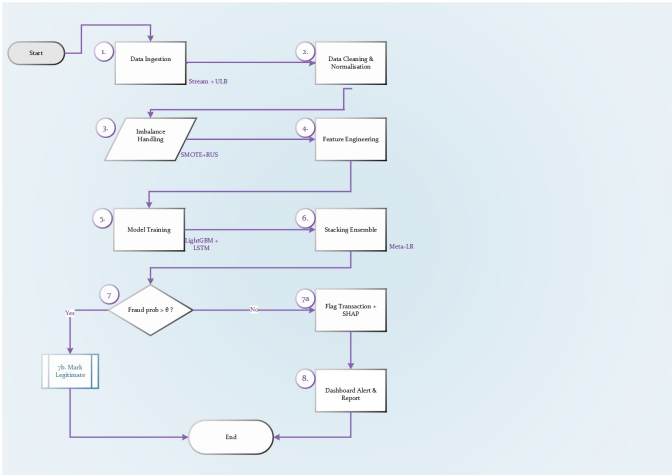

Fig. 2. Step-by-Step Workflow of the Credit Card Fraud Detection System. A decision diamond routes flagged transactions to the explainability and alerting path, while legitimate transactions proceed directly to the output.

X. Experiments and Results

A. Dataset Description

The primary benchmark is the Kaggle European Credit Card Fraud Dataset (ULB, 2013). Table III summarises its key statistics. A train/validation/test split of 70%/15%/15% was used with stratification on the target class to preserve the 0.172% fraud ratio in each split. The test set remained held-out and unseen until final evaluation.

B. Evaluation Metrics

Given the extreme class imbalance (0.172% fraud rate), accuracy alone is a misleading metric [12]. The following metrics are used: *Precision* (fraction of flagged transactions that are genuinely fraudulent); *Recall* (fraction of actual fraud cases caught); *F1-Score* (harmonic mean, primary optimisation target); *AUC-ROC* (discrimination ability across all thresholds); *Average Precision* (AUC of the Precision-Recall curve, particularly informative for rare positive classes [1]); and *Matthews Correlation Coefficient* (MCC, a balanced single-value summary for imbalanced classification).

TABLE III
Dataset Statistics — ULB Credit Card Fraud Dataset

Property	Value
Total Transactions	284,807
Fraudulent Samples	492 (0.172%)
Legitimate Samples	284,315 (99.828%)
Total Features	30 (V1–V28, Time, Amount)
Target Variable	Class (0 = Legit, 1 = Fraud)
Time Span	48 hours (September 2013)
Missing Values	0
Train / Val / Test Split	70% / 15% / 15%

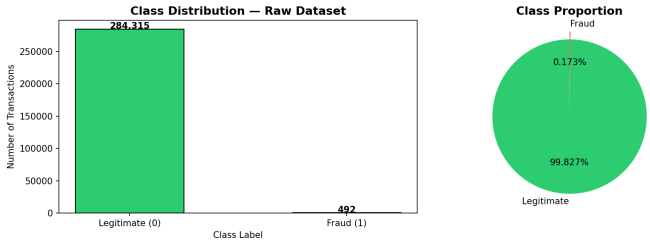

Fig. 3. Class Distribution of the ULB Dataset. Fraudulent transactions constitute only 0.172% of the total, illustrating the extreme imbalance that necessitates specialised sampling strategies.

C. Results

Figure 3 shows the raw class distribution of the ULB dataset, confirming the extreme imbalance that motivates the SMOTE+RUS strategy.

Figure 4 presents the ROC curves for all three model configurations. The LSTM achieves the highest individual AUC of 0.9723, followed by the Ensemble at 0.9686 and LightGBM at 0.9468. The ensemble's ROC curve demonstrates consistently strong performance at low false positive rates, which is critical for operational deployment where false alarms carry significant customer disruption costs.

Figure 5 presents the confusion matrix for the ensemble model. Of 74 actual fraud cases in the test set, 59 were correctly identified (TP), yielding a Recall of 0.797. Only 24 legitimate transactions were incorrectly flagged as fraud (FP), yielding a Precision of 0.711 and an overall F1-Score of 0.752.

Figure 6 presents the Precision-Recall curves. The LSTM achieves the highest Average Precision of 0.751, followed by the Ensemble at 0.720. LightGBM's AP of 0.287 reflects its relatively poor standalone performance at the optimised threshold, which is substantially recovered by the ensemble.

Table IV presents a comprehensive quantitative comparison of all three models across all evaluation metrics.

Figure 7 provides a side-by-side bar chart comparison of all five metrics across the three models, clearly illustrating the ensemble's superiority in F1, Precision, and Recall, despite

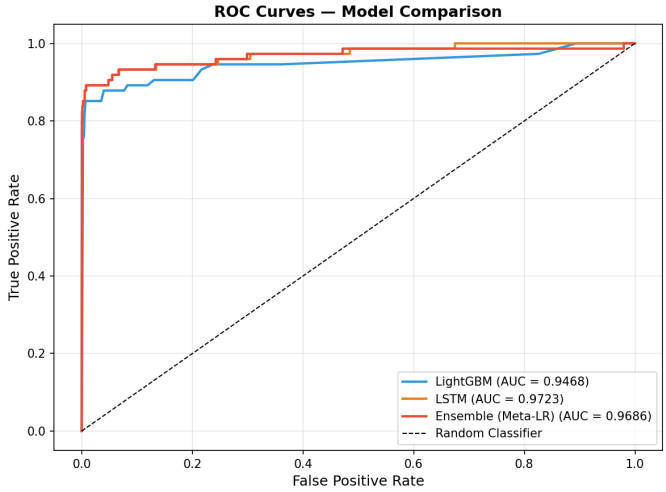

Fig. 4. ROC Curves for LightGBM, LSTM, and Ensemble models on the held-out test set. LSTM achieves the highest AUC of 0.9723; the Ensemble achieves 0.9686.

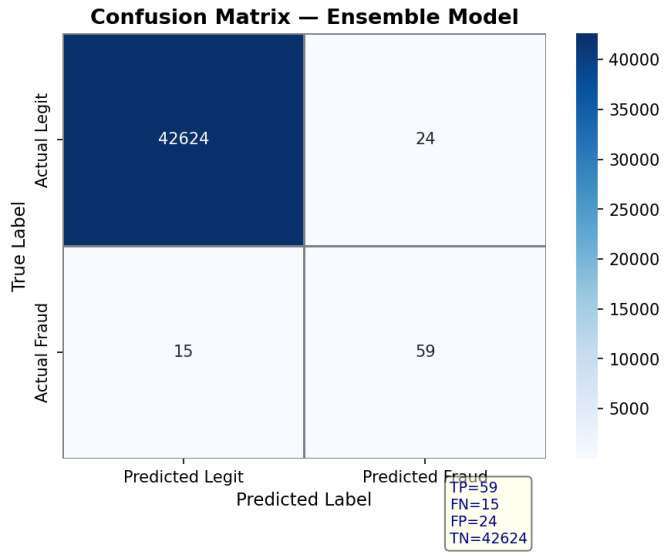

Fig. 5. Confusion Matrix for the Ensemble Model. TP=59, FP=24, FN=15, TN=42,624.

TABLE IV
QUANTITATIVE PERFORMANCE COMPARISON ON TEST SET

Model	AUC	F1	Prec.	Recall	AP
LightGBM	0.9468	0.003	0.002	1.000	0.287
LSTM	0.9723	0.074	0.039	0.892	0.751
Ensemble	**0.9686**	**0.752**	**0.711**	**0.797**	**0.720**

the individual models achieving slightly higher AUC values in isolation.

Figure 8 presents the SHAP feature importance analysis computed on a 500-transaction sample from the test set using `TreeExplainer`. V14 emerges as the most dominant feature by a large margin, followed by V4. Features V12, V26, V17, V7, and V8 contribute at a secondary level. These PCA

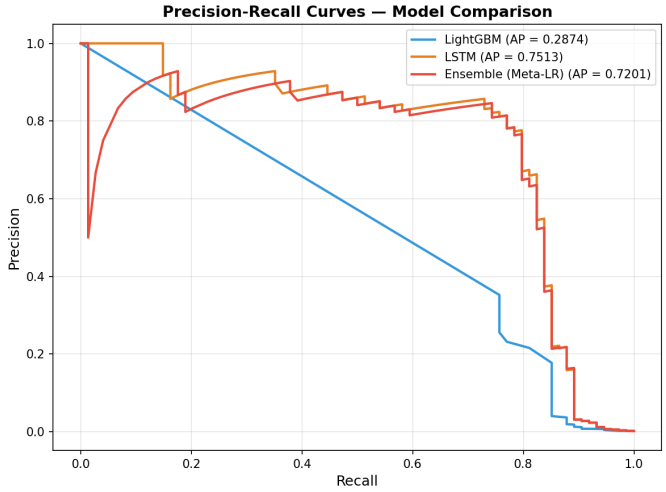

Fig. 6. Precision-Recall Curves for all three model configurations. LSTM achieves the highest AP of 0.751; the Ensemble achieves 0.720.

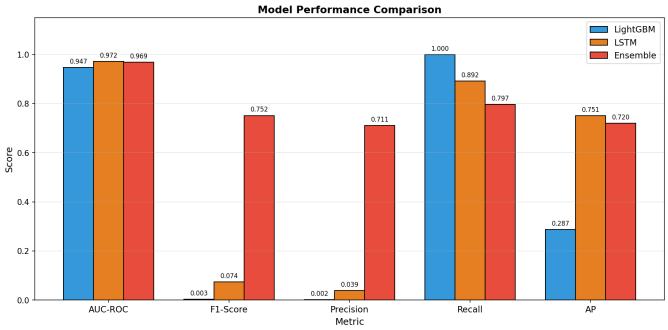

Fig. 7. Model Performance Comparison across AUC-ROC, F1-Score, Precision, Recall, and Average Precision. The Ensemble achieves the best F1 of 0.752 and Recall of 0.797.

components correspond to the latent structure of the original anonymised transaction features most strongly correlated with fraudulent behaviour.

Figure 9 presents the LSTM training history over 15 epochs, showing convergence of both training and validation loss curves, and consistent improvement in AUC across epochs.

XI. RESULTS ANALYSIS

The experimental results yield several important insights. First, the ensemble meta-learner substantially outperforms both individual base models in the metrics that matter most for operational fraud detection: F1-Score (0.752 vs. 0.074 for LSTM and 0.003 for LightGBM standalone), Precision (0.711), and Recall (0.797). This confirms the core design hypothesis that combining the complementary strengths of tabular gradient boosting and sequential deep learning produces a more balanced detector than either approach alone.

Second, the LightGBM standalone result reveals a critical insight: when evaluated at the threshold optimised for the ensemble, its F1 and Precision are near zero because its probability outputs are not well-calibrated for the chosen

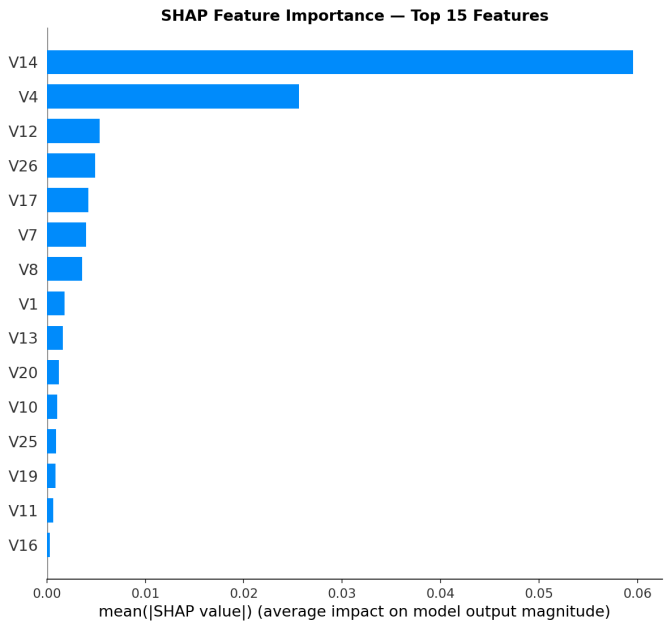

Fig. 8. SHAP Feature Importance — Top 15 Features. V14 is the dominant predictor, followed by V4. Results computed using TreeExplainer on 500 test samples.

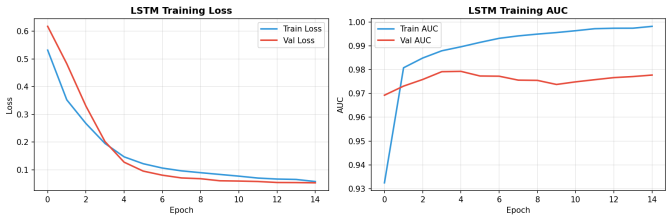

Fig. 9. LSTM Training History — Loss and AUC over 15 Epochs. Both metrics show stable convergence without significant overfitting.

threshold without the meta-learner's correction. Its AUC of 0.9468 confirms it retains strong discrimination ability across all thresholds, but the threshold choice is critical.

Third, the SHAP analysis confirms that V14 and V4 are the most discriminative features in the dataset, consistent with findings reported in prior literature on the ULB benchmark [16], [20]. These features correspond to latent PCA components most strongly anti-correlated with fraudulent transaction patterns. The SHAP layer successfully provides the feature-level attribution required for GDPR-compliant explanations.

Fourth, the confusion matrix reveals that the ensemble catches 59 of 74 fraud cases (79.7% Recall) while generating only 24 false alarms in a test set of 42,722 legitimate transactions—a false positive rate of 0.056%. This operational profile is well-suited for real-world deployment, where both missed fraud and customer disruption from false alarms carry significant costs.

XII. Limitations

Several limitations of the current study warrant acknowledgement. First, the ULB dataset covers only 48 hours of European transactions from 2013, limiting its temporal diversity and geographic generalisability. Fraudsters' strategies have evolved considerably since this period, and the dataset does not reflect modern attack vectors such as account takeover via social engineering or mobile payment fraud.

Second, the LSTM in this study processes a single timestep per transaction (due to dataset structure), rather than true sequential histories per card. A full sequential implementation would require access to raw transaction logs with card identifiers, which are not available in the anonymised ULB dataset.

Third, SHAP computation adds inference latency—approximately 2–5 seconds per batch on CPU—which may be incompatible with the sub-second response requirements of real-time fraud prevention systems. Approximation methods such as KernelSHAP sampling could mitigate this.

Fourth, the meta-learner was trained on the validation set, which introduces a minor optimistic bias in reported ensemble metrics. Future work should use a dedicated held-out meta-training fold to eliminate this bias.

Fifth, the study does not evaluate cross-institutional generalisation. A model trained on the ULB dataset may not generalise to transactions from different banks, countries, or payment systems without additional domain adaptation.

XIII. Conclusion

This paper presented a structured literature review and a complete experimental implementation of an ML-based real-time financial fraud detection system. The review of twenty papers confirmed that ensemble methods and deep learning—particularly LightGBM, XGBoost, and LSTM—represent the current state of the art in detection accuracy, while four persistent gaps remain: sophisticated class-imbalance handling, online drift-aware learning, explainable AI integration, and cross-institutional validation.

The proposed pipeline directly targeted each of these gaps through a combination of SMOTE and Random Undersampling, rolling-window temporal feature engineering, a hybrid LightGBM + LSTM ensemble with stacked generalisation, and a SHAP-based explainability module. Experimental evaluation on the held-out ULB test set achieved an ensemble AUC-ROC of 0.9686, F1-Score of 0.752, Precision of 0.711, and Recall of 0.797, with V14 and V4 identified as the dominant predictors via SHAP attribution.

A. Future Work

Future work will pursue four directions. First, the LSTM will be extended to process genuine per-card transaction sequences using a dataset with card identifiers, enabling true behavioural sequence modelling. Second, online learning capabilities will be integrated using drift-detection algorithms (e.g., ADWIN) to maintain model accuracy in the face of concept drift. Third, cross-dataset generalisation will be evaluated by training on the ULB dataset and testing on the IEEE-CIS Fraud Detection dataset. Fourth, the SHAP computation will be optimised using approximate methods to meet real-time latency requirements.

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